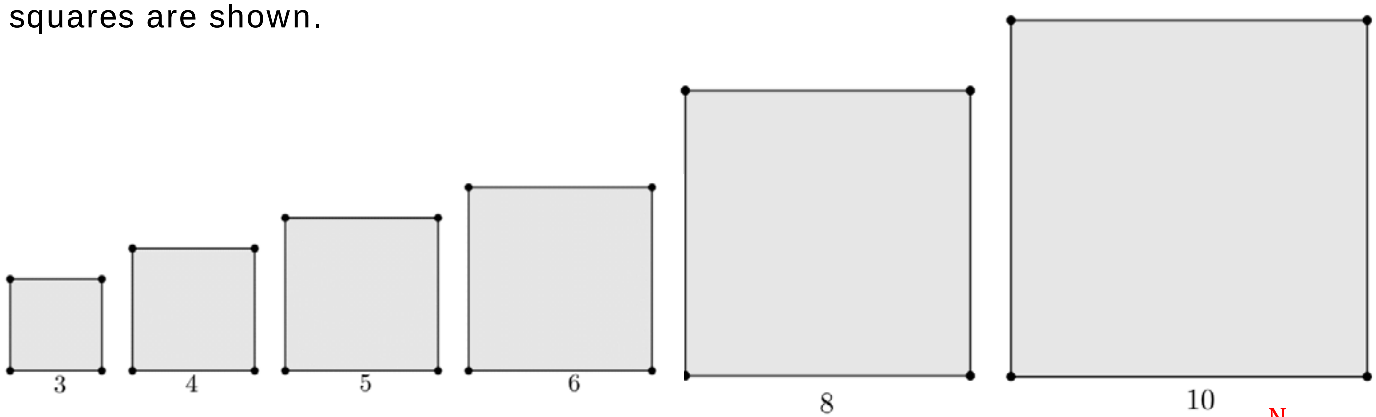
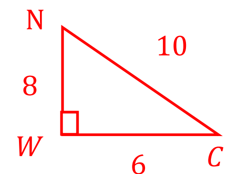
Geometry
Unit 7
Lesson 1 Practice

Name_Answer Key________
Date___________________
Period_________________

Six squares are shown.



The side measures of right triangle NWC are: $WC = 6$, $WN = 8$, and $NC = 10$.

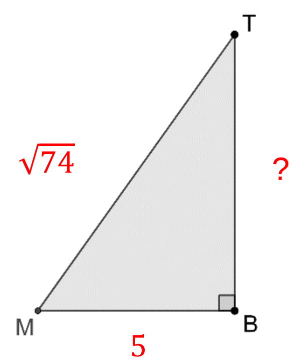


1. To show a visual proof of the Pythagorean theorem which square, if any, should be placed along $\overline{WN}$?
   8×8 square

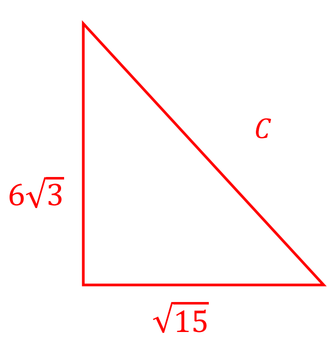
2. To show a visual proof of the Pythagorean theorem which square, if any, should be placed along $\overline{WC}$?
   6×6 square

3. $\triangle MTB$ is shown where $MT = \sqrt{74}$ and $MB = 5$.
   Find the exact length of $\overline{BT}$.

$$a^2 + b^2 = c^2$$
$$5^2 + b^2 = \left(\sqrt{74}\right)^2$$
$$25 + b^2 = 74$$
$$\underline{-25 \qquad -25}$$
$$\sqrt{b^2} = \sqrt{49}$$
$$b = 7$$
$$\overline{BT} = 7$$



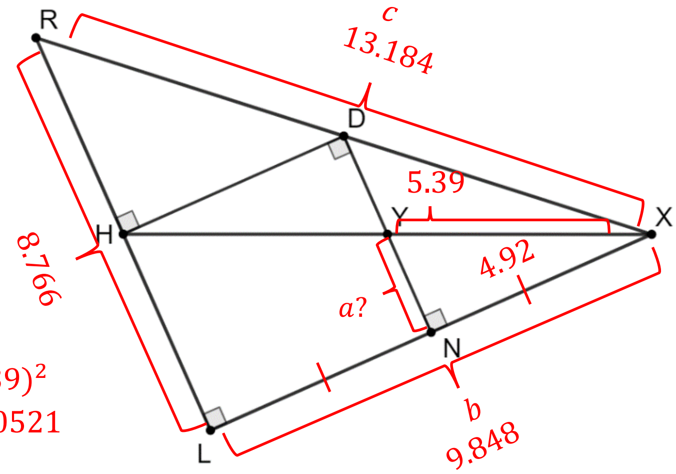
4. A right triangle has side lengths of $6\sqrt{3}$ and $\sqrt{15}$.
   Find the exact length of the hypotenuse.



$$a^2 + b^2 = c^2$$
$$\left(6\sqrt{3}\right)^2 + \left(\sqrt{15}\right)^2 = c^2$$
$$108 + 15 = c^2$$
$$\sqrt{123} = \sqrt{c^2}$$
$$c = \sqrt{123}$$

5. Triangle RLX is shown where $LN = NX$, $RL = 8.766$, $XR = 13.184$, and $YX = 5.39$.

Determine the length of $\overline{NY}$. If necessary, round your answer to the nearest thousandth.



1) find length of $\overline{LX}$

$(8.766)^2 + b^2 = (13.184)^2$

$76.842756 + b^2 = 173.817856$

$b^2 = \sqrt{96.9751}$

$b \approx 9.848$

$NX = 4.924$

2) Find $\overline{YN}$

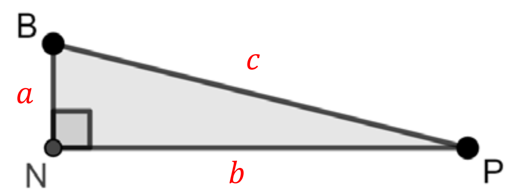
$(4.924)^2 + a^2 = (5.39)^2$

$24.2458 + a^2 = 29.0521$

$\sqrt{a^2} = \sqrt{4.806324}$

$a \approx 2.192$ $\quad YN = 2.192$

$\triangle BNP$ is shown. Complete the table by finding the missing lengths. Round your answer to the nearest thousandth. Show your work below the table. $a^2 + b^2 = c^2$



	Length of $\overline{BN}$ a leg	Length of $\overline{NP}$ b leg	Length of $\overline{BP}$ c hypotenuse
6.	48	55	73
7.	4.123	5	$\sqrt{42}$
8.	7	$\sqrt{95}$	12
9.	33	56	65
10.	5.196	$\sqrt{2}$	$\sqrt{29}$

6. $48^2 + b^2 = 73^2$

$2304 + b^2 = 5329$

$\sqrt{b^2} = \sqrt{3025}$

$b = 55$

7. $a^2 + 5^2 = (\sqrt{42})^2$

$a^2 + 25 = 42$

$\sqrt{a^2} = \sqrt{17}$

$a \approx 4.123$

8. $7^2 + (\sqrt{95})^2 = c^2$

$49 + 95 = c^2$

$\sqrt{c^2} = \sqrt{144}$

$c = 12$

9. $33^2 + b^2 = 65^2$

$1089 + b^2 = 4225$

$\sqrt{b^2} = \sqrt{3136}$

$b \approx 56$

10. $a^2 + (\sqrt{2})^2 = (\sqrt{29})^2$

$a^2 + 2 = 29$

$\sqrt{a^2} = \sqrt{27}$

$a \approx 5.196$